VERIFICATION OF THE WEIGHTED SZEGED TREE-MINIMIZER CONJECTURE THROUGH ORDER ELEVEN

Abstract. Conjecture 2 of Bok, Furtula, Jedličková, and Škrekovski asserts that, among connected graphs of a fixed order, a tree attains the minimum weighted Szeged index. For each $3 \leq n \leq 11$, we evaluate the invariant on one representative of every isomorphism class of connected simple n-vertex graph, totaling 1,018,690,327 graphs. The conjecture holds throughout this range. At $n = 3$, the path P_3 and the triangle K_3 tie at value 12; for every $4 \leq n \leq 11$, every minimizer is a tree. We also determine the exact minimum over non-trees at each order. This is a finite verification and does not resolve the conjecture for $n \geq 12$.

1. Result

All graphs are finite, simple, and connected. For an edge $e = uv$ of G, set

$$n_u(e) = \big|\{x \in V(G) : d_G(x,u) < d_G(x,v)\}\big|,$$

and define $n_v(e)$ analogously. The weighted Szeged index is

$$\mathrm{wSz}(G) = \sum_{uv \in E(G)} (\deg_G(u) + \deg_G(v)) n_u(uv) n_v(uv).$$

Vertices equidistant from the endpoints of an edge are counted in neither factor. The index was introduced by Ilić and Milosavljević [1]. Bok et al. [2] posed the following conjecture.

Weighted Szeged tree-minimizer conjecture [2, Conjecture 2]**.** For n-vertex graphs the minimum weighted Szeged index is attained by a tree.

The displayed sentence is the source's wording. Connectivity is implicit in it: $n_u(e)$ is defined through graph distances, and the computer tests reported in [2] range over connected graphs, so we read the statement over connected graphs and take every minimum below over that class. The source poses two conjectures. Its Conjecture 1, that the *maximum* weighted Szeged index over connected graphs of a given order is attained by the balanced complete bipartite graph, was proved after submission by Cambie [3], as a remark in the published version of [2] records. The minimization statement displayed above is the one at issue here, and it is the one that remains open.

The cases $n = 1, 2$ are immediate, so the census begins at $n = 3$, the first order admitting a connected non-tree. Let $\mathcal{G}_n$ be the set of isomorphism

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classes of connected simple graphs on n vertices, let $\mathcal{T}_n \subseteq \mathcal{G}_n$ be the trees, and let $\mathcal{N}_n = \mathcal{G}_n \setminus \mathcal{T}_n$.

Theorem 1. *For every* $3 \leq n \leq 11$*,*

$$\min_{G \in \mathcal{G}_n} \mathrm{wSz}(G) = \min_{T \in \mathcal{T}_n} \mathrm{wSz}(T).$$

At $n = 3$*, both* P_3 *and* K_3 *attain the minimum* 12*. For every* $4 \leq n \leq 11$*, every minimizer is a tree. The exact tree and non-tree minima are given in Table 1.*

TABLE 1. Minimum weighted Szeged indices from the exhaustive census.

n	connected classes	tree minimum	non-tree minimum	gap
3	2	12	12	0
4	6	34	36	2
5	21	72	76	4
6	112	130	132	2
7	853	204	212	8
8	11,117	306	316	10
9	261,080	432	444	12
10	11,716,571	578	600	22
11	1,006,700,565	762	778	16

The tree minima agree with those reported by Hell, Hernández-Cruz, and Hosseini [4]. The census additionally determines the exact minimum over non-trees throughout the completed range.

2. Exhaustive computation

We generated one representative of every isomorphism class in $\mathcal{G}_n$ with `geng` from the nauty and Traces package [5]. For each graph, breadth-first search from every vertex produced the distance matrix. The defining sum was then evaluated in exact integer arithmetic. Because every generated graph was connected, it was classified as a tree precisely when it had $n-1$ edges.

The order-10 census was completed as a single stream, as 37 disjoint edge-count shards, and with residue-class shardings modulo 64 and 256. The order-11 census was repeated with residue-class shardings modulo 240 and 512. The complete runs used two separately written graph6 decoders and two distance kernels, repeated breadth-first search and Floyd–Warshall. All runs returned the same graph counts, minimum values, and displayed witnesses.

As external checks, the numbers of connected graphs and trees agree with OEIS A001349 and A000055, respectively. At order 10, the 37 edge-count shard sizes agree term by term with OEIS A054924 [6]. The evaluator also reproduces the published tree minima for every $3 \le n \le 11$.

A further check compares the objects rather than the values. At the orders $7 \le n \le 11$, where our range meets the minimum-wSz trees published in [2, Table 1], the tree witnesses of Table 2 are isomorphic to the trees displayed there: at $n = 7, 8, 9$ they are the spiders $S(2,2,2)$, $S(2,2,3)$ and $S(2,2,2,2)$, and at $n = 10, 11$ each consists of two adjacent vertices of degree 3 carrying legs of length 2, with a single leg of length 3 at $n = 11$. These structures are readable directly from the graph6 strings.

Proof of Theorem 1. The census evaluates one representative of every isomorphism class in $\mathcal{G}_n$, and wSz is invariant under isomorphism. Hence the two restricted minima are the entries of Table 1. Direct evaluation gives

$$\mathrm{wSz}(P_3) = \mathrm{wSz}(K_3) = 12.$$

For $4 \le n \le 11$, the minimum over $\mathcal{N}_n$ is strictly larger than the minimum over $\mathcal{T}_n$, so every global minimizer is a tree. $\square$

3. Witnesses and scope

Table 2 gives a graph6 witness for the tree minimum at every completed order, and for the non-tree minimum at the two largest orders. These records certify the displayed upper bounds directly from the definition; optimality follows from the exhaustive census. At $n = 6$ the tree minimum is attained by two isomorphism classes and one of them is listed. Every string in the table was re-decoded, re-encoded byte for byte, and re-evaluated by both distance kernels.

Table 2. Graph6 witnesses attaining the minima of Table 1.

n	class	graph6	wSz
3	tree	`Bo`	12
4	tree	`Cq`	34
5	tree	`DkG`	72
6	tree	`Eh_O`	130
7	tree	`FCOf?`	204
8	tree	`GiE@?O`	306
9	tree	`HiQ@?_G`	432
10	tree	`I?AA@_gw?`	578
10	non-tree	`I?AA@_gwO`	600
11	tree	`J?AA@AOEB_?`	762
11	non-tree	`J?AACGoIF??`	778

Theorem 1 is a finite computer-assisted verification. It does not settle the conjecture for $n \ge 12$. Bok et al. publish the minimum-wSz trees for the orders 7 to 25 [2, Table 1]; the computer tests supporting Conjecture 2, namely those ranging over all connected graphs, are reported without stating the orders covered, and that census is not published. The statement was still recorded as open after those tests: Hell, Hernández-Cruz, and Hosseini restate it as their Conjecture 1.1 and describe characterizing the graphs of fixed order with minimum weighted Szeged index as one of the main open problems in the area [4].

Exact verification. Every number printed above that can be recomputed from the data of this note has been recomputed independently, by a second program written from the statements here rather than from the census code, using a standard library only and exact integer arithmetic. It re-derives the four $n = 10$ and $n = 11$ witnesses of Table 2 and their indices by two distance kernels and a third edge-split kernel for trees, together with their byte-exact graph6 re-encodings; the tie $\mathrm{wSz}(P_3) = \mathrm{wSz}(K_3) = 12$ at $n = 3$; the whole tree-minimum column for $3 \leq n \leq 11$, by exhaustive free-tree enumeration; both minima for $3 \leq n \leq 9$, by exhaustive generation for $n \leq 8$ and a covering enumeration at $n = 9$; the connected-class column and its total 1,018,690,327, from a cycle-index count and the inverse Euler transform independently of OEIS; and the gap column. It reports 56 checks and all 56 pass. Its non-tree search at $n = 10$ and $n = 11$ was exhaustive only over the connected graphs with at most 13 edges, where it returns 600 and 778, and the bound $\mathrm{wSz}(G) \geq \sum_v \deg(v)^2$ closes $m \geq 39$ at $n = 10$ and $m \geq 47$ at $n = 11$; the intermediate edge counts require `geng`, and the shard replications were not repeated, so the non-tree minima at those two orders, the last two gaps, and the conclusion that every minimizer is a tree there rest on the census reported above. The program and its transcript are retained in an auxiliary archive available on request.

Reproducibility

The census uses nothing beyond `geng` of [5] and an evaluator of the defining sum, so the exact invocations and the evaluator are printed here in full. One representative of every isomorphism class in $\mathcal{G}_n$ is generated as a single stream, as one shard per edge count, and as residue shards, by

```
geng -c -q  n       # every connected class on n vertices
geng -c -q  n  m:m  # one shard per edge count m,
                    #   m = n-1,...,n(n-1)/2
                    #   (37 shards at n = 10)
geng -c -q  n  r/s  # one residue shard r = 0,...,s-1,
                    #   s = 64, 256  (n = 10)
                    #   s = 240, 512 (n = 11)
```

and each graph6 line of the output is evaluated by

```
read n and the adjacency of the graph from the graph6 line
for s in 0..n-1:  d[s][*] <- BFS(s)   # or Floyd-Warshall
w <- 0
for each edge uv:
    nu <- #{x : d[x][u] < d[x][v]}
```

```
    nv <- #{x : d[x][v] < d[x][u]}
    w  <- w + (deg(u) + deg(v)) * nu * nv
if (number of edges) = n-1:
    update (tree minimum, witness) with (w, line)
else:
    update (non-tree minimum, witness) with (w, line)
```

All arithmetic is exact integer arithmetic; there is no randomization, no floating point, and no tolerance parameter. Every entry of Tables 1 and 2 is an output of this loop, and each witness can be rechecked against the definition from its graph6 string alone. No data file or program accompanies this note, and none is needed to recheck it: the generated shard files are too large to deposit and are regenerated by the commands above rather than reproduced, and the independent program of Section 3 takes no input beyond the tables printed here.

References

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